Zero-Touch Provisioning of a Branch Edge (NCX-1000)

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Summary

This article describes end-to-end zero-touch provisioning (ZTP) for onboarding an NCX-1000 branch edge into NetCore SD-WAN 5.0 using NetCore Orchestrator. It covers prerequisites, device template preparation, serial allow-list registration, bootstrap behavior, verification, rollback, and common ZTP failures including NCS-ZTP-4001, NCS-ZTP-4003, and NCS-ORC-6001.

Applies to: NetCore SD-WAN 5.0, NetCore Orchestrator, NCX-1000 branch edge

Overview

Zero-touch provisioning allows an NCX-1000 branch edge to be shipped directly to a site, powered on, connected to a transport, and automatically onboarded into NetCore Orchestrator without manual site configuration on the device.

In NetCore SD-WAN 5.0, the ZTP flow requires a valid device certificate, an approved serial number in the Orchestrator allow-list, a matching device template, and successful control-plane connectivity after bootstrap. If the redirect or certificate validation phase fails, Orchestrator raises ZTP alarms. Use KB-NET-1008 for the full NCS alarm and error code reference.

- The NCX-1000 obtains initial network reachability through the connected transport, typically Internet broadband, MPLS, or LTE/5G.
- The device contacts the NetCore ZTP redirect service and is redirected to the assigned NetCore Orchestrator.
- NetCore Orchestrator validates the device certificate and serial number.
- The assigned branch template is pushed to the NCX-1000.
- The NCX-1000 establishes control connections to Orchestrator and begins SD-WAN operation.

ZTP-related alarms covered in this article

Code	Meaning	Typical cause during onboarding
NCS-ZTP-4001	ZTP Redirect Failed	Device cannot reach the redirect service, redirect record is missing, or the serial number is not associated with the correct Orchestrator tenant
NCS-ZTP-4003	Device Certificate Validation Failed	Device certificate is expired, not trusted, mismatched to the serial number, or the appliance date/time is incorrect
NCS-ORC-6001	Orchestrator Control Connection Lost	Device completed bootstrap but cannot maintain management-plane connectivity to NetCore Orchestrator

Prerequisites

Complete these checks before shipping or powering on the NCX-1000 at the branch. Missing prerequisites are the most common cause of failed ZTP onboarding.

- NetCore Orchestrator is running NetCore SD-WAN 5.0 and is reachable from the branch transport network.
- The NCX-1000 serial number is available from the device label or shipment manifest. Example serial: NCX1000-BR-7K4D92.
- The NCX-1000 is factory defaulted or has no conflicting local startup configuration.
- At least one branch transport is available: MPLS, Internet broadband, or LTE/5G.

- The branch access circuit provides IP reachability for initial onboarding. If static addressing is required at the site, prepare the bootstrap values before dispatch.
- The device certificate has not expired and matches the NCX-1000 serial number.
- Firewall policy allows the NCX-1000 to reach NetCore Orchestrator and the NetCore ZTP redirect service.
- A site object, device template, and serial allow-list entry exist in NetCore Orchestrator before the appliance boots onsite.
- Operations staff have access to the NCX-1000 console in case a local ZTP restart is required with request sdwan ztp restart.

Step-by-step Procedure

Perform the procedure in the order shown. The examples below use site BR-021 in Dallas, an NCX-1000 serial number of NCX1000-BR-7K4D92, and NetCore Orchestrator tenant Enterprise-WAN.

- 1. Create or confirm the branch site in NetCore Orchestrator. Use a consistent site ID, location, and role so that the correct branch policies are applied.
- 2. Create the NCX-1000 device template. Include the device model, site binding, system profile, transport interfaces, and SD-WAN role.
- 3. Add the NCX-1000 serial number to the serial allow-list and bind it to the site and device template.
- 4. Confirm that the ZTP redirect target is the correct NetCore Orchestrator tenant. A wrong tenant or missing redirect record can raise NCS-ZTP-4001.
- 5. Ship or install the NCX-1000 at the branch. Connect the transport cables before applying power. For a typical branch, connect ge0/0 to Internet broadband and ge0/1 to MPLS if available.
- 6. Power on the NCX-1000. Do not manually configure SD-WAN policy on the device during the first boot.
- 7. If the site requires local bootstrap for static addressing, connect to the console and apply only the minimum transport reachability settings. Then restart ZTP.
- 8. Monitor the onboarding job in NetCore Orchestrator. The expected state sequence is Pending Redirect, Certificate Validated, Template Assigned, Config Pushed, Control Connected, and In Service.
- 9. If onboarding stalls after template push, verify the control connection. A post-bootstrap control-plane loss is reported as NCS-ORC-6001.

```
# Example: create the site, template, and serial allow-list entry by using the NetCore Orchestrator API
# Replace ORC_TOKEN and orchestrator address with values for your tenant.

curl -sk -H "Authorization: Bearer ORC_TOKEN" \
  -H "Content-Type: application/json" \
  -X POST https://orchestrator.netcore.example/api/v5/sites \
  -d '{
    "siteId": "BR-021",
    "name": "Dallas-Branch-021",
    "role": "branch",
    "timezone": "America/Chicago"
  }'

curl -sk -H "Authorization: Bearer ORC_TOKEN" \
  -H "Content-Type: application/json" \
  -X POST https://orchestrator.netcore.example/api/v5/device-templates \
  -d '{
    "templateName": "TPL-NCX1000-BRANCH-DUAL-TRANSPORT",
    "model": "NCX-1000",
    "release": "5.0",
    "role": "branch-edge",
    "interfaces": [
      {"name": "ge0/0", "transport": "Internet broadband", "addressing": "dhcp", "enabled": true},
      {"name": "ge0/1", "transport": "MPLS", "addressing": "dhcp", "enabled": true},
      {"name": "cell0/0", "transport": "LTE/5G", "addressing": "dhcp", "enabled": false}
    ],
    "controlPolicy": "default-branch-control",
    "qosPolicy": "branch-standard-qos"
  }'
curl -sk -H "Authorization: Bearer ORC_TOKEN" \
```

```
  -H "Content-Type: application/json" \
  -X POST https://orchestrator.netcore.example/api/v5/ztp/allow-list \
  -d '{
    "serialNumber": "NCX1000-BR-7K4D92",
    "model": "NCX-1000",
    "siteId": "BR-021",
    "templateName": "TPL-NCX1000-BRANCH-DUAL-TRANSPORT",
    "redirectTenant": "Enterprise-WAN",
    "ztpAction": "auto-onboard"
  }'

# Optional local bootstrap when the branch transport does not provide DHCP.
# Apply only initial reachability settings, then restart ZTP.
ncx-edge# configure terminal
ncx-edge(config)# interface ge0/0
ncx-edge(config-if)# description INTERNET-BROADBAND-ZTP
ncx-edge(config-if)# ip address 198.51.100.42/29
ncx-edge(config-if)# no shutdown
ncx-edge(config-if)# exit
ncx-edge(config)# ip route 0.0.0.0/0 198.51.100.41
ncx-edge(config)# commit
ncx-edge(config)# exit
ncx-edge# request sdwan ztp restart
```

Verification

After the Orchestrator onboarding job reports In Service, verify from the NCX-1000 CLI that the device has an active control connection and that BFD sessions are established for SD-WAN tunnel monitoring. The examples below show a healthy dual-transport branch edge.

If ZTP fails, correlate the Orchestrator onboarding job with the alarm code. For detailed alarm definitions and remediation paths, see KB-NET-1008.

- Verify control-plane connectivity to NetCore Orchestrator with show sdwan control connections.
- Verify tunnel health with show sdwan bfd sessions.
- Verify that SLA classes were downloaded with show sdwan sla-class.
- If application routing is enabled for the branch, verify path measurements with show sdwan app-route statistics.

Common ZTP failure indicators

Symptom	Alarm code	Immediate checks
Device remains in Pending Redirect or Redirect Failed	NCS-ZTP-4001	Confirm branch reachability to the ZTP redirect service, verify the serial number NCX1000-BR-7K4D92 is allow-listed, and confirm the redirect tenant is Enterprise-WAN
Device reaches Orchestrator but is rejected during authentication	NCS-ZTP-4003	Confirm the device certificate matches the serial number, check appliance date/time, and verify the serial entry has not been assigned to another device
Device receives template but later appears offline in Orchestrator	NCS-ORC-6001	Verify transport reachability, default route, branch firewall policy, and control connection status on the NCX-1000

```
ncx-edge# show sdwan control connections
TIME                 PEER                          STATE        ROLE          UPTIME     LAST-ERROR
2026-02-18 14:22:31  orchestrator.netcore.local    connected    orchestrator  00:18:44   none

ncx-edge# show sdwan bfd sessions
SESSION-ID  LOCAL-IF  TRANSPORT             REMOTE-EDGE  STATE  LATENCY  JITTER  LOSS
1021        ge0/0     Internet broadband    NCX-4000-RH1  up     42 ms    6 ms    0.1%
1022        ge0/1     MPLS                  NCX-4000-RH1  up     18 ms    2 ms    0.0%

ncx-edge# show sdwan sla-class
CLASS               LATENCY-THRESHOLD  JITTER-THRESHOLD  LOSS-THRESHOLD
Voice               150 ms             30 ms             1%
Video               200 ms             40 ms             2%
```

```
Business-Critical   250 ms            50 ms            3%
Best-Effort         400 ms            n/a              5%

ncx-edge# show sdwan app-route statistics
APP-CLASS           SELECTED-TRANSPORT    LATENCY  JITTER  LOSS  SLA-STATE
Voice               MPLS                  18 ms    2 ms    0.0%  compliant
Video               MPLS                  18 ms    2 ms    0.0%  compliant
Business-Critical   MPLS                  18 ms    2 ms    0.0%  compliant
Best-Effort         Internet broadband    42 ms    n/a     0.1%  compliant
```

Rollback

Use rollback when the wrong template, site, or serial number was assigned, or when a staged NCX-1000 must be removed from service before it completes onboarding. Rollback removes the Orchestrator assignment and returns the device to a state where ZTP can be attempted again with the corrected allow-list and template.

If the device has already established a control connection, remove or disable the device object from NetCore Orchestrator before restarting ZTP locally. If the device never completed redirect or certificate validation, correct the Orchestrator records first and then restart ZTP.

- 1. In NetCore Orchestrator, disable the device object for serial number NCX1000-BR-7K4D92.
- 2. Remove the serial allow-list entry or reassign it to the correct site and template.
- 3. If a wrong template was pushed, replace it with the intended NCX-1000 template before retrying ZTP.
- 4. On the NCX-1000 console, restart ZTP after Orchestrator has been corrected.
- 5. Re-run verification commands after the device returns to In Service.

```
# Disable the incorrect device object
curl -sk -H "Authorization: Bearer ORC_TOKEN" \
  -H "Content-Type: application/json" \
  -X PATCH https://orchestrator.netcore.example/api/v5/devices/NCX1000-BR-7K4D92 \
  -d '{"adminState":"disabled"}'

# Remove the incorrect allow-list binding
curl -sk -H "Authorization: Bearer ORC_TOKEN" \
  -X DELETE https://orchestrator.netcore.example/api/v5/ztp/allow-list/NCX1000-BR-7K4D92

# Restart ZTP from the local console after correcting Orchestrator records
ncx-edge# request sdwan ztp restart

# Confirm control connectivity after retry
ncx-edge# show sdwan control connections
```

Known Limitations

The following limitations apply to NCX-1000 ZTP onboarding in NetCore SD-WAN 5.0.

- ZTP requires the NCX-1000 serial number to be present in the NetCore Orchestrator allow-list before the device attempts onboarding.
- An NCX-1000 with an invalid or mismatched device certificate cannot be onboarded through ZTP. The expected alarm is NCS-ZTP-4003.
- If the device cannot reach the ZTP redirect service or the redirect record points to the wrong tenant, onboarding fails with NCS-ZTP-4001.
- If the NCX-1000 completes bootstrap but loses the Orchestrator control connection, Orchestrator reports NCS-ORC-6001. Resolve control-plane reachability before troubleshooting application routing.
- Do not use an NCX-4000 or NCX-9000 template for an NCX-1000 branch edge. The model in the template must be NCX-1000.
- Only minimal local bootstrap should be applied before ZTP. Full SD-WAN policy should be delivered by NetCore Orchestrator.
- For additional alarm descriptions and event correlation guidance, see KB-NET-1008.

Related Articles

KB-NET-1008: NCS Alarm and Error Code Reference